Notes: Pomeranz (AER, 2015)

No Taxation without Information: Deterrence and Self-Enforcement in the Value Added Tax

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1 Big picture

- **Question.** Does the value added tax (VAT) really help tax enforcement by creating third-party paper trails between firms? The paper asks whether these paper trails deter evasion ex ante and whether enforcement on one firm spills over to its trading partners.
- **Setting.** The setting is Chile, where the VAT has a single 19 percent rate and is paid monthly. Firms remit VAT on sales minus input costs and must keep original receipts to document both tax debits and tax credits.
- **Main economic mechanism.** A buyer firm wants a receipt because input costs reduce its VAT bill. That receipt is also a record of the seller's sale. The buyer therefore becomes a third-party information provider about the seller.
- **Where the mechanism breaks.** The paper trail is strongest for business-to-business transactions. It weakens at the retail stage because final consumers do not need receipts to deduct input costs from a VAT bill.
- **Experiment 1: Letter Message Experiment.** The Chilean Tax Authority randomly sends deterrence letters to over 100,000 firms, raising the perceived probability of audit. The paper compares tax responses where the VAT paper trail is present versus absent.
- **Experiment 2: Spillover Experiment.** A separate sample of 5,600 micro firms suspected of evasion is randomly assigned to receive an audit preannouncement. The tax authority later collects information about their suppliers and clients to test whether enforcement propagates through the VAT chain.
- **Core result 1.** The deterrence letter strongly increases VAT payments, but the response is much weaker for transactions covered by a paper trail than for sales to final consumers.
- **Core result 2.** The audit preannouncement creates strong spillovers to suppliers of treated firms, but not to clients. This directionality is exactly what the VAT self-enforcement mechanism predicts.
- **Interpretation.** For an individual firm, the VAT paper trail and the firm's own audit probability act as substitutes: if a transaction already has a paper trail, an additional perceived audit risk creates less incremental response. For the system as a whole, the paper trail and audit probability act as complements because one firm's monitoring can raise compliance by other firms.
- **Economic takeaway.** Tax instruments that are equivalent in textbook models can be very different once evasion and information flows matter. The VAT is valuable not only because of its tax base, but because it generates verifiable third-party information.

2 Incremental contribution of the paper

- **From theory to micro-experimental evidence.** The self-enforcement argument for the VAT is old and influential, but before this paper there was little direct micro evidence showing that the VAT paper trail actually changes firm compliance.
- **Direct evidence on third-party information among firms.** Previous evidence on third-party reporting often focuses on individual income taxes. This paper studies firms and inter-firm

transactions, where firms act as fiscal intermediaries and information aggregators.

- **A large-scale randomized policy test.** The Letter Message Experiment covers hundreds of thousands of firms in an economy with a fully functioning VAT system. This lets the paper test the paper trail in general equilibrium, not only in a small pilot sample.
- **A mechanism experiment on supply-chain spillovers.** The Spillover Experiment shows the mechanism in action by increasing deterrence for one firm and then measuring tax responses by its suppliers and clients.
- **Distinction between preventive deterrence and ex post audit detection.** A paper trail makes evasion easier to detect during an audit. But the paper asks whether firms anticipate that detectability and therefore evade less before any audit occurs.
- **Within-firm transaction comparison.** The paper compares different line items within the same firm. This helps separate the effect of the paper trail from firm-level characteristics such as size, risk aversion, tax morale, or managerial ability.
- **Directionality of spillovers.** The paper does not only ask whether enforcement spills over; it asks whether spillovers move up the VAT chain, as the theory predicts. Suppliers respond strongly; clients do not.
- **Audit-strategy contribution.** The results show that a tax authority may want to consider indirect effects through trading networks when choosing whom to audit.
- **Developing-country relevance.** The paper studies Chile, a middle-income country with a strong tax authority, and speaks to the broader challenge of building enforceable tax systems where many transactions are hard to observe.
- **Tax-instrument contribution.** The paper demonstrates that the administrative and informational features of tax instruments can matter as much as statutory rates and formal tax bases.

3 Institutional context and VAT mechanism

3.1 The Chilean VAT

- Chile uses a single 19 percent VAT rate.
- Firms pay VAT monthly on declared sales minus declared input costs.
- In simple notation, firm i's VAT liability is roughly

$$\mathrm{VAT}_i = \tau \left(\widehat{s}_i - \widehat{c}_i \right),$$

  where τ is the VAT rate, $\widehat{s}_i$ is reported sales, and $\widehat{c}_i$ is reported input costs.
- The VAT base is equivalent to a final sales tax base when there is a uniform rate and no exemptions, but the remittance and information structure is different.
- Firms must document their reported tax debit and tax credit with original receipts. Receipts are approved and stamped by the tax authority.

- The tax authority can cross-check a buyer's reported input costs against a seller's reported sales, but for many firms this information is accessible mainly during an audit rather than automatically in electronic form.

Interpretation: The VAT is not automatically self-enforcing just because receipts exist. The paper trail matters only when firms expect the government may use it.

3.2 Why the VAT creates a paper trail

- If firm i sells to firm $i+1$, the buyer wants a receipt to claim the input-cost deduction.
- The same receipt reveals a sale by firm i and can therefore be used by the tax authority to verify the seller's declaration.
- Buyer and seller have opposing incentives on the reported amount: the buyer prefers high input costs, while the seller prefers low sales.
- This makes collusion difficult in the middle of the chain.
- The mechanism breaks down at the final stage because consumers do not have a VAT deduction motive.

Interpretation: The VAT uses the incentives of firms against one another. The buyer's desire to reduce its own tax liability creates information that can discipline the seller.

3.3 Types of evasion

- **Unilateral evasion** occurs when the seller's and buyer's records do not match. For example, the seller underreports sales while the buyer still reports the input receipt.
- **Collusive evasion** occurs when both parties omit or understate a transaction consistently in both sets of books.
- Collusive evasion is more plausible if it begins at the final stage, where the consumer has no deduction incentive, and then propagates upstream.
- An increase in the audit probability for a firm should increase reported VAT for its suppliers if suppliers must report more sales to match the treated firm's records.
- The effect on clients is ambiguous because higher reported sales by the treated firm may let the client claim more input costs.

Interpretation: The sign and direction of spillovers are informative. If suppliers respond but clients do not, that pattern is not just general fear of the tax authority; it is evidence of the VAT chain mechanism.

4 Data and measurement

4.1 Letter Message Experiment sample

- The sample contains 445,734 small and medium-size firms with valid postal addresses.
- The sample excludes VAT-exempt firms, very large firms, firms in the Spillover Experiment, and firms without valid postal addresses.

- The deterrence letter is sent to 102,031 firms in two waves: 84,618 firms in the first wave and 17,413 firms in the second wave.
- The tax morale letter is sent to 18,579 firms.
- The placebo letter is sent to 18,519 firms.
- The control group contains 306,605 firms that receive no letter.
- The main outcomes are monthly VAT filings.

Interpretation: This experiment is large enough to study heterogeneous responses by transaction type and firm type. It is especially useful for testing whether paper-trail transactions respond differently to deterrence.

4.2 Spillover Experiment sample

- The sample contains 5,600 micro-size firms suspected of evasion.
- The firms are mostly rural, often remote, and frequently report input costs larger than sales.
- Half of the firms are randomly assigned to receive an audit preannouncement; the other half receive no letter.
- All 5,600 firms are later summoned for audit.
- Because the tax authority does not generally know firm trading partners, auditors collect pre-treatment information about suppliers and clients during the audits.
- The final audited sample contains 1,527 firms, and the tax authority obtains information about 2,829 trading partners.

Interpretation: The Spillover Experiment intentionally focuses on a low-compliance environment. This gives the paper a chance to observe the VAT paper trail being activated by stronger deterrence.

4.3 Outcome variables

- The main outcome is monthly declared VAT, excluding carryovers from earlier periods so that the analysis focuses on current-month transactions.
- Because declared VAT has extreme dispersion and many observations at zero, the paper uses median regressions and binary outcomes in addition to means.
- The main binary outcomes are whether declared VAT is greater than zero, greater than the same month of the previous year, and greater than a predicted benchmark based on the control group.
- For transaction-level analysis, the paper examines total sales, input costs, intermediary sales to other firms, and final sales to consumers.

Interpretation: The distribution of VAT declarations is very skewed. Median and threshold-based outcomes are therefore more informative than only studying mean pesos.

5 Models and empirical specifications

5.1 Difference-in-differences for the Letter Message Experiment

The core quantile specification is

$$Q_\tau(VAT_{it} \mid Z_i, t) = \alpha_\tau + \beta_\tau(Z_i \times Post_t) + \gamma_\tau Z_i + \delta_t,$$

where Z_i is treatment assignment and $Post_t$ indicates the post-treatment period.

- β_τ measures the treatment effect at quantile τ, especially the median.
- Month fixed effects control for aggregate time patterns in tax filings.
- The design compares treated firms with control firms before and after the letter.

The linear probability specification is

$$T_{it} = \alpha + \beta(Z_i \times Post_t) + \gamma Z_i + \delta_t + e_{it},$$

where T_{it} is an indicator such as VAT greater than prior-year VAT or VAT greater than predicted VAT.

Interpretation: These specifications translate the randomized letter into a change in perceived audit risk. Changes in VAT declarations are interpreted as reductions in evasion, holding real activity fixed in the short run.

5.2 Within-firm line-item specification

To compare different types of transactions within the same firm, the paper estimates specifications of the form

$$\begin{aligned} LineItemIncrease_{itl} = \alpha + \beta(c_i \times Post_t) + c_i + \delta_t \\ + \phi(Z_i \times Post_t \times l) + \eta(Post_t \times l) + \iota_l + \varepsilon_{itl}. \end{aligned}$$

- l indexes the line item, such as sales, input costs, intermediary sales, or final sales.
- Firm fixed effects c_i absorb permanent firm differences.
- Firm-by-post controls absorb firm-specific post-treatment shifts.
- The coefficient ϕ is the differential treatment effect for the line item.

Interpretation: This is a powerful check because it compares different transactions within the same firm. The concern that retail firms and intermediary firms differ in many ways is reduced.

5.3 Spillover specification

For trading partners, the treatment is whether their audited trading partner received the audit preannouncement. The core regression compares suppliers and clients of treated versus control firms before and after the preannouncement.

- The main dependent variables are again whether VAT is higher than in the previous year and whether VAT is higher than predicted.

- Supplier and client indicators allow the paper to test the direction of spillovers.
- Standard errors are clustered at the level of the audited firm.
- Controls include firm sales, sales-to-input ratio, share of final sales, and whether the industry is categorized as hard to monitor.

Interpretation: A supplier response but not a client response is the signature prediction of the VAT self-enforcement mechanism.

6 Identification and empirical design

6.1 What the experiments identify

- The Letter Message Experiment identifies the causal effect of increasing perceived audit probability on tax declarations.
- The key test for the paper trail is not the overall effect of the letter. It is whether the response differs between transactions with and without a VAT paper trail.
- The Spillover Experiment identifies whether a randomly induced increase in monitoring for one firm affects its trading partners.
- The supplier-client comparison identifies whether spillovers move in the direction predicted by the VAT chain.

Interpretation: The paper uses randomized variation in deterrence to study a non-randomized institution: the VAT paper trail. This is an indirect policy evaluation of the VAT's enforcement mechanism.

6.2 Why the deterrence letter is interpreted as deterrence

- The deterrence letter tells firms they were selected for analysis and could be audited if irregularities are detected.
- The content was pretested with firm representatives, who interpreted it as raising the probability of audit.
- The placebo letter controls for receiving a letter from the tax authority.
- The tax morale letter controls for a motivational appeal to social norms.
- Only the deterrence letter generates the immediate, large response in VAT payments.

Interpretation: The comparison across letter types rules out a simple attention or mail-from-the-government channel. The response is driven by perceived enforcement risk.

6.3 Balance and attrition

- In the Letter Message Experiment, treatment and control groups are balanced on pretreatment VAT, firm age, non-filing, size, retail status, intermediary status, and final-sales share.
- In the Spillover Experiment, the full treatment and control samples are balanced before treatment.

- Only about 27 percent of the Spillover Experiment firms are actually audited because of administrative delays and a major earthquake, but audited treatment and control firms remain largely balanced.
- Trading partners of treated and control firms are also balanced on most observable characteristics.

Interpretation: Attrition is the main challenge in the Spillover Experiment. The paper argues that attrition is plausibly unrelated to treatment, and balance tables support this claim.

6.4 Main assumptions and limitations

- In the Letter Message Experiment, the inference that lower response implies lower ex ante evasion assumes that no other within-firm difference between final sales and inter-firm transactions drives the differential response.
- In the Spillover Experiment, the tax authority observes trading partners only for audited firms, so the analysis depends on the audited subsample.
- Some spillovers to the control group may exist, especially in the large Letter Message Experiment. These would tend to attenuate estimated effects.
- Short-run increases in declared VAT could partly reflect real activity changes, but the paper argues that the immediate response is best interpreted as reduced evasion.
- Chile has a relatively capable tax authority. Generalization to countries with weaker enforcement or more corruption may require caution.

Interpretation: The design is strong for detecting paper-trail mechanisms, but the paper does not claim that VAT self-enforcement works automatically in every institutional environment.

7 Tables and figures

7.1 Table 1: Responses to Increase in Audit Probability: Collusive and Unilateral Evasion

Read:

- Table 1 summarizes the theoretical predictions for suppliers, the treated firm, and clients when the treated firm's audit probability increases.
- Under collusive evasion, a treated firm that starts reporting transactions can force its suppliers to report more sales, raising supplier VAT.
- The treated firm may report both higher inputs and higher sales. Its VAT effect is therefore positive or ambiguous depending on relative changes.
- Clients may be able to claim more input costs if the treated firm reports more sales, so client VAT can fall under collusive evasion.
- Under unilateral evasion, discrepancies are corrected, and the predicted VAT response is positive for suppliers, the treated firm, and clients.

Economic message: The table gives the logic behind the Spillover Experiment. A supplier response larger than a client response is not just a generic fear effect; it is the expected sign of the VAT paper trail moving up the production chain.

7.2 Figure 1: Research Design Diagrams

Read:

- Figure 1 lays out the two randomized experiments.
- The left panel shows the Letter Message Experiment among 445,734 small and medium-size firms: deterrence letter, tax morale letter, placebo letter, and no-letter control group.
- The right panel shows the Spillover Experiment among 5,600 micro firms suspected of evasion, randomly split between audit preannouncement and control.
- The figure shows that trading partner information is collected after audits, using records from before treatment.

Economic message: The paper combines a broad economy-wide deterrence experiment with a targeted mechanism experiment. The first tests whether the paper trail already deters evasion; the second shows how deterrence propagates along the chain.

Table 1—Responses to Increase in Audit Probability: Collusive and Unilateral Evasion

Position in supply chain	Collusive evasion		Unilateral evasion	
Supplier	Sales ↑	VAT ↑	Sales ↑	VAT ↑
Treated firm	Inputs ↑ Sales ↑	VAT (↑)	Inputs ↓ Sales ↑	VAT ↑
Client	Inputs ↑	VAT ↓	Inputs ↓	VAT ↑

Notes: "Collusive evasion" stands for the type of evasion where a transaction is omitted from the books of both the seller and the buyer firm. "Unilateral evasion" stands for the type of evasion where the books of the seller and the buyer reveal discrepancies. Buyers, for whom inputs represent a tax deduction, will tend to overstate the value of the transaction, while sellers, for whom the transaction represents a tax liability, will tend to understate its value. The arrows indicate the expected direction of change for the line item in question resulting from an increased audit probability on the treated firm.

Table 1: Responses to Increase in Audit Probability: Collusive and Unilateral Evasion

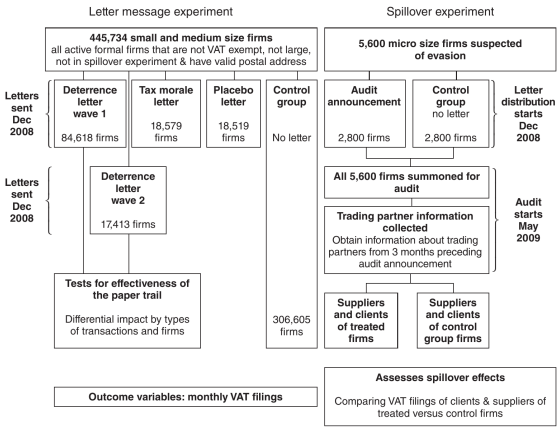

Figure 1. Research Design Diagrams

Figure 1: Research Design Diagrams

7.3 Table 2: Letter Message Experiment: Baseline Summary Statistics and Balance of Randomization

Read:

- Table 2 compares the control group to the deterrence, tax morale, and placebo treatment groups before treatment.
- The control group has mean monthly VAT of 264,029 Chilean pesos and median monthly VAT of 17,511 pesos.

- Micro firms make up about 74.5 percent of the control group; small firms 18.2 percent; medium-size firms 2.8 percent.
- Retail firms account for 28.7 percent and intermediary firms 38.2 percent.
- None of the differences between the treatment groups and the control group are statistically significant at the 10 percent level.

Economic message: The balance table supports the randomized design. Pretreatment differences in VAT, size, and position in the production chain cannot explain later treatment effects.

7.4 Table 3: Spillover Experiment: Baseline Summary Statistics and Balance of Randomization

Read:

- Table 3 reports balance for the full Spillover Experiment sample, the audited subsample, and the trading partners of audited firms.
- The full sample is highly unusual: mean monthly VAT is negative, because these firms often declare input costs greater than sales.
- Treatment and control firms are balanced in the full sample.
- Among audited firms, treatment and control are also mostly balanced, although the treated audited group has a somewhat higher agriculture share.
- Trading partners of treated and control firms are balanced on most variables; trading partners are much larger than the audited firms.
- The trading-partner sample includes both suppliers and clients, with suppliers accounting for about 56.5 percent of trading partners.

Economic message: The table shows why the Spillover Experiment is useful: it focuses on firms with suspiciously low VAT declarations, but the randomization remains balanced enough to compare the trading partners of treated and control firms.

Table 2—Letter Message Experiment: Baseline Summary Statistics and Balance of Randomization

	Control group (1)	Difference to deterrence (2)	Difference to tax morale (3)	Difference to placebo (4)
Monthly VAT (mean)	264,029	3,105	305	−10,565
	(1,871)	(3,744)	(7,869)	(7,472)
Monthly VAT (median)	17,511	−188	−415	−7
	(115)	(231)	(491)	(462)
Firm age in months	108	−0.32	−0.43	−0.60
	(0.12)	(0.25)	(0.51)	(0.51)
Percent non-filed declarations	4.3	−0.03	−0.07	−0.02
	(0.02)	(0.05)	(0.10)	(0.10)
Percent no sales year prior	1.5	−0.05	0.06	−0.01
	(0.02)	(0.04)	(0.09)	(0.09)
Percent micro size	74.5	−0.04	−0.28	−0.06
	(0.08)	(0.16)	(0.33)	(0.33)
Percent small size	18.2	0.028	0.102	−0.055
	(0.07)	(0.14)	(0.29)	(0.29)
Percent medium size	2.8	0.01	0.02	0.07
	(0.03)	(0.06)	(0.12)	(0.13)
Percent retail firms	28.7	−0.06	−0.28	−0.13
	(0.08)	(0.16)	(0.34)	(0.34)
Percent intermediary firms	38.2	0.15	0.31	−0.08
	(0.09)	(0.18)	(0.37)	(0.37)
Percent final sales	45.6	−0.10	−0.29	−0.07
	(0.08)	(0.17)	(0.35)	(0.35)
Number of firms	306,605	102,031	18,579	18,519

Notes: Each row shows a regression of the pretreatment variable in question on treatment dummies and a constant term. The constant term captures the value for the control group. Monthly VAT winzorized at the top and bottom 0.1 percent to deal with extreme outliers. Columns 2–4 show the difference of the treatment groups to the control group. None of the differences are statistically significant at the 10 percent-level. Monetary amounts are in Chilean pesos, with 500 pesos approximately equivalent to US$1. Robust standard errors in parentheses, clustered at the firm level for all variables except for median tax paid, for which the table shows the result of a median regression for October 2008, the month before the tax payment. All other observations are monthly for ten months prior to

Table 2: Letter Message Experiment: Baseline Summary Statistics and Balance of Randomization

Table 3—Spillover Experiment: Baseline Summary Statistics and Balance of Randomization

	All firms		Audited firms		Trading partners	
	Control group (1)	Difference (2)	Control group (3)	Difference (4)	Trading partners of control group (5)	Difference (6)
Monthly VAT (mean)	−18,452	−3,849	−8,024	−16,385	1,069,113	29,274
	(11,370)	(12,326)	(6,271)	(12,988)	(82,548)	(114,952)
Monthly VAT (median)	0	0	0	0	301,230	10,750
	(0)	(0)	(0)	(0)	(24,656)	(34,396)
Firm age in months	131	−0.43	139	−3.33	126	2.01
	(1.19)	(1.69)	(2.18)	(3.13)	(1.77)	(2.51)
Percent non-filed declaration	1.09	0.1	0.40	−0.21	1.31	−0.43*
	(0.13)	(0.18)	(0.11)	(0.13)	(1.94)	(0.24)
Size category	2.28	−0.00	2.34	0.02	5.87	0.065
	(0.01)	(0.01)	(0.02)	(0.03)	(0.62)	(0.9)
Percent retail firms	6.82	−0.54	7.27	−1.16	2.26	−0.63
	(0.48)	(0.66)	(0.94)	(1.28)	(0.49)	(0.6)
Percent intermediary firms	66.3	−0.46	63	1.01	35.7	−2.37
	(0.89)	(1.27)	(1.76)	(2.46)	(1.52)	(2.13)
Percent final sales	15.7	−0.35	19.4	−2.99	22.7	0.15
	(0.65)	(0.91)	(1.35)	(1.82)	(1.1)	(1.46)
Sales/input	0.68	−0.01	0.73	−0.03	1.64	−0.15
	(0.01)	(0.01)	(0.02)	(0.02)	(0.1)	(0.11)
Percent agriculture	54	0.71	45	5.34**	18.2	1.72
	(0.94)	(1.33)	(1.81)	(2.55)	(1.32)	(1.84)
Percent audited	27	0.46	100	0		
	(0.84)	(1.19)	(.)	(.)		
Percent suppliers					56.5	−0.0032
					(1.46)	(2.11)
Number of firms	2,800	2,800	757	770	1,444	1,385

Notes: This table shows summary statistics for the pretreatment period and balance of randomization for three groups: the 5,600 firms in the sample of the Spillover Experiment, the firms that were actually audited, and the trading partners of the audited firms. Each row shows three regressions of the pretreatment variable in question on a dummy indicating treatment assignment and a constant term: columns 1 and 2 for the firms in the full Spillover Experiment sample, columns 3 and 4 for the audited firms, and columns 5 and 6 for the trading partners. Observations are monthly for ten months prior to treatment (January 2008–October 2008), except for median tax paid, for which observations are averaged over ten months prior to treatment. The constant terms shown in columns 1, 3, and 5 capture the values for the control group. Columns 2, 4, and 6 show the difference of the treatment group to the control group. Monetary amounts are in Chilean pesos, with 500 Chilean pesos approximately equiva-

Table 3: Spillover Experiment: Baseline Summary Statistics and Balance of Randomization

7.5 Figure 2: Impact of the Three Types of Letters

Read:

- Panel A shows a clear jump in median VAT payments after the deterrence letter is mailed.
- The median VAT response rises by roughly 10 to 12 percent shortly after treatment, then gradually decays.
- Panels B and C show no comparable immediate jump for the tax morale and placebo letters.
- The timing is important: the deterrence effect appears immediately after the letter, consistent with a change in perceived audit risk.

Economic message: The deterrence letter changes behavior because it changes expected enforcement. A general motivational appeal or a placebo contact from the tax authority does not produce the same immediate compliance response.

7.6 Table 4: Letter Message Experiment: Intent-to-Treat Effects on VAT Payments by Type of Letter

Read:

- Table 4 quantifies the letter effects using several outcomes.
- The deterrence letter increases median VAT by 1,326 pesos per month, significant at the 1 percent level.

- It raises the probability of declaring more VAT than in the previous year by 1.40 percentage points.
- It raises the probability of declaring more than predicted by 1.42 percentage points.
- It raises the probability of declaring positive VAT by 0.53 percentage points.
- The tax morale letter has little effect except on the probability of declaring positive VAT; the placebo letter has no significant effect.
- Mean VAT is noisy and not statistically significant, consistent with the extremely skewed distribution of tax declarations.

Economic message: The deterrence treatment works, and it works immediately. This validates the letter as a tool for raising perceived audit probability, which is the key experimental shock used to study the VAT paper trail.

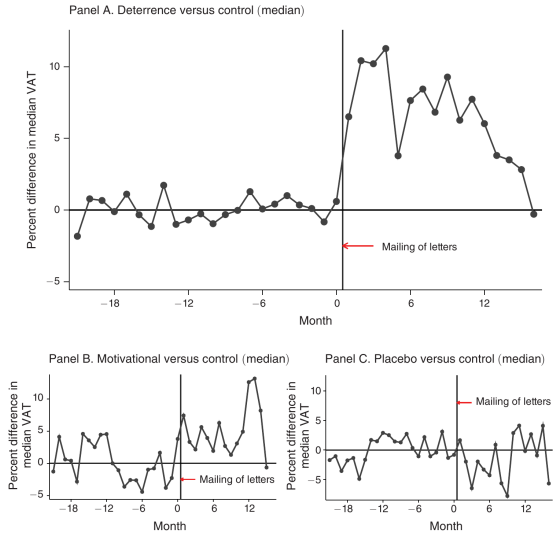

FIGURE 2. IMPACT OF THE THREE TYPES OF LETTERS

Notes: This figure plots the monthly percent difference between the medians of the treatment and the control group for each type of letter: (median VAT treatment group—median VAT control group)/(median VAT control group), normalizing the average of pretreatment months percent difference to zero. The *y*-axis indicates time, with monthly observations, and zero indicates the last month before the mailing of the letters. The vertical line marks mailing of the letters. The figure shows the first wave of mailing. For the second (much smaller) wave of mailing, see online Appendix Figure A2.

Figure 2: Impact of the Three Types of Letters

TABLE 4—LETTER MESSAGE EXPERIMENT: INTENT-TO-TREAT EFFECTS ON VAT PAYMENTS BY TYPE OF LETTER

	Mean VAT (1)	Median VAT (2)	Percent VAT > previous year (3)	Percent VAT > predicted (4)	Percent VAT > zero (5)
Deterrence letter × post	−1,114	1,326***	1.40***	1.42***	0.53***
	(2,804)	(316)	(0.12)	(0.10)	(0.09)
Tax morale letter × post	−1,840	262	0.40	0.30	0.44**
	(6,082)	(666)	(0.25)	(0.22)	(0.20)
Placebo letter × post	835	383	−0.11	−0.19	−0.14
	(6,243)	(687)	(0.26)	(0.23)	(0.20)
Constant	268,810***	17,518***	47.50***	48.27***	67.30***
	(1,799)	(112)	(0.07)	(0.07)	(0.06)
Month fixed effects	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	No	Yes	Yes	Yes
Treatment assignment	No	Yes	No	No	No
Observations	7,892,076	1,221,828	7,892,076	7,892,076	7,892,076
Number of firms	445,734	445,734	445,734	445,734	445,734
Adjusted R^2	0.40		0.14	0.28	0.47

Notes: Column 1 shows a regression of the mean declared VAT on treatment dummies, winsorized at the top and bottom 0.1 percent to deal with extreme outliers. Column 2 shows a median regression of average VAT before treatment and in four months after each treatment wave. Columns 3–5 show linear probability regressions of the probability of an increase in declared VAT compared to the same month in the previous year, the probability of declaring more than predicted and the probability of declaring any positive amount. Observations are monthly in columns 1 and 3–5 for ten months prior to treatment and four months after each wave of mailing. The four months after the second wave excludes firms treated in the first. Coefficients and standard errors of the linear probability regressions are multiplied by 100 to express effects in percent. Monetary amounts are in Chilean pesos, with 500 Chilean pesos approximately equivalent to $1. Standard errors in parentheses, robust and clustered at the firm level for columns 1 and 3–5.

*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

Table 4: Letter Message Experiment: Intent-to-Treat Effects on VAT Payments by Type of Letter

7.7 Table 5: Impact of Deterrence Letter on Different Types of Transactions

Read:

- Table 5 compares the deterrence response across transaction types among firms with both final and intermediary sales.
- Total sales respond strongly: the probability that sales exceed the same month of the previous year rises by 1.17 percentage points.

- Input costs, which are based on inter-firm transactions and are covered by the paper trail, do not respond significantly.
- Intermediary sales to other firms do not respond significantly.
- Final sales to consumers, where the VAT paper trail is absent, rise strongly by 1.33 percentage points.

Economic message: This is the central paper-trail result. Firms respond to deterrence mainly where third-party paper trails are weak. Where the VAT already creates paper-trail information, additional deterrence has much less incremental effect.

7.8 Table 6: Interaction of Firm Size and Share of Sales to Final Consumers

Read:

- Table 6 tests whether the deterrence response varies with the share of final sales and with firm size.
- Firms with a larger share of final sales respond more strongly to the deterrence letter.
- Smaller firms respond more than larger firms, consistent with the idea that large firms have more internal and external monitoring.
- Once the share of final sales is controlled for, the negative relationship between firm size and response becomes much smaller.
- In the specification using VAT greater than predicted, the size interaction becomes statistically insignificant after controlling for final-sales share.

Economic message: Some of the higher evasion response among small firms appears to reflect the fact that small firms are more likely to sell to final consumers, where the VAT paper trail is weakest.

Table 5—Impact of Deterrence Letter on Different Types of Transactions

	Percent sales > previous year (1)	Percent input costs > previous year (2)	Percent intermediary sales > previous year (3)	Percent final sales > previous year (4)
Deterrence letter × post	1.17***	0.16	0.12	1.33***
	(0.22)	(0.21)	(0.19)	(0.21)
Constant	55.39***	53.25***	38.37***	45.04***
	(0.13)	(0.13)	(0.12)	(0.12)
Month fixed effects	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Observations	2,392,529	2,392,529	2,392,529	2,392,529
Number of firms	133,156	133,156	133,156	133,156
Adjusted R^2	0.25	0.22	0.30	0.32

Notes: Regressions of the probability of the line item (total sales, total input costs, intermediary sales, and final sales) being higher than in the same month the previous year. Sample of firms that have both final and intermediary sales in the year prior to treatment. The four months after the second wave excludes firms treated in the first wave. Coefficients and standard errors are multiplied by 100 to express effects in percent. Robust standard errors in parentheses, clustered at the firm level.

*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

Table 5: Impact of Deterrence Letter on Different Types of Transactions

Table 6—Interaction of Firm Size and Share of Sales to Final Consumers

	Percent VAT > previous year				
	(1)	(2)	(3)	(4)	(5)
Panel A					
Deterrence letter × final sales share	1.61***			1.48***	1.43***
	(0.26)			(0.27)	(0.26)
Deterrence letter × size category		−0.17***		−0.10***	
		(0.04)		(0.04)	
Deterrence letter × log employees			−0.45***		−0.29**
			(0.11)		(0.12)
Deterrence letter	0.68***	2.63***	1.66***	1.49***	0.92***
	(0.16)	(0.29)	(0.13)	(0.35)	(0.19)
Constant	47.53***	48.87***	47.50***	48.89***	47.53***
	(0.08)	(0.08)	(0.08)	(0.08)	(0.08)
Final sales share × post	Yes	No	No	Yes	Yes
Size measure × post	No	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Month dummies	Yes	Yes	Yes	Yes	Yes
Observations	7,308,631	7,116,590	7,340,994	7,084,823	7,308,631
Number of firms	406,834	396,135	408,636	394,367	406,834
Adjusted R^2	0.14	0.14	0.14	0.14	0.14
	Percent VAT > predicted				
	(1)	(2)	(3)	(4)	(5)
Panel B					
Deterrence letter × final sales share	1.51***			1.51***	1.44***
	(0.23)			(0.25)	(0.24)
Deterrence letter × size category		−0.10***		−0.03	
		(0.03)		(0.04)	
Deterrence letter × log employees			−0.28***		−0.11
			(0.10)		(0.11)
Deterrence letter	0.74***	2.15***	1.57***	1.00***	0.83***
	(0.14)	(0.26)	(0.12)	(0.32)	(0.16)
Constant	48.48***	49.79***	48.26***	50.01***	48.48***
	(0.08)	(0.08)	(0.08)	(0.08)	(0.08)
Final sales share × post	Yes	No	No	Yes	Yes
Size measure × post	No	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	7,308,631	7,116,590	7,340,994	7,084,823	7,308,631
Number of firms	406,834	396,135	408,636	394,367	406,834
Adjusted R^2	0.28	0.26	0.29	0.26	0.28

Notes: Regression of the probability of monthly declared VAT being higher than in the same month of the previous year (panel A) and on being higher than predicted (panel B). Coefficients and standard errors are multiplied by 100 to express effects in percent. Sample includes all firms in the deterrence treatment and in the control group. The four months after the second wave excludes firms treated in the first. Number of observations vary due to missing observations for some variables. Final sales share is not defined for firms with zero sales in preceding year, size category is not available for new firms. Robust standard errors in parentheses, clustered at the firm level.

*** Significant at the 1 percent level.
** Significant at the 5 percent level.

Table 6: Interaction of Firm Size and Share of Sales to Final Consumers

7.9 Table 7: Spillover Effects on Trading Partners' VAT Payments

Read:

- Table 7 tests whether an audit preannouncement sent to one firm affects the VAT payments of its trading partners.
- For all trading partners combined, the audit preannouncement increases the probability of VAT exceeding the previous year by 2.41 percentage points.
- The effect is concentrated among suppliers: supplier VAT increases by 4.28 percentage points relative to the previous-year benchmark and 3.92 percentage points relative to the predicted benchmark.
- Client effects are small and statistically insignificant.
- The supplier effect remains strong after controlling for firm sales, sales-to-input ratio, final-sales share, and hard-to-monitor industry status.

Economic message: The supplier-client asymmetry is the direct mechanism evidence. Enforcement on one firm causes upstream suppliers to report more VAT, showing that the paper trail transmits deterrence through the production chain.

Table 7—Spillover Effects on Trading Partners' VAT Payments

	Percent VAT > previous year (1)	Percent VAT > predicted (2)	Percent VAT > previous year (3)	Percent VAT > predicted (4)	Percent VAT > previous year (5)	Percent VAT > predicted (6)
Audit announcement × post	2.41**	2.03*				
	(1.14)	(1.11)				
Audit announcement × supplier × post			4.28***	3.92***	4.14***	3.83***
			(1.54)	(1.50)	(1.52)	(1.52)
Audit announcement × client × post			−0.26	−0.28	−0.14	−0.28
			(1.64)	(1.51)	(1.67)	(1.55)
Supplier × post			−0.64	0.34	−1.11	0.60
			(1.62)	(1.59)	(1.67)	(1.64)
Constant	52.07***	49.06***	52.07***	49.06***	52.75***	50.11***
	(0.95)	(0.94)	(0.95)	(0.94)	(0.96)	(0.96)
Controls × post	No	No	No	No	Yes	Yes
Controls × audit announcement × post	No	No	No	No	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	45,264	45,264	45,264	45,264	44,288	44,288
Number of firms	2,829	2,829	2,829	2,829	2,768	2,768
Adjusted R^2	0.05	0.11	0.05	0.11	0.05	0.10

Notes: Regressions for trading partners of audited firms. Columns 1, 3, and 5 show the probability of an increase in declared VAT since the previous year, columns 2, 4, and 6 show the probability of declaring more than predicted. The controls in columns 5 and 6 are firm sales, sales/input-ratio, share of sales going to final consumers, and industry categorized as "hard-to-monitor." Observations are monthly for ten months prior to treatment and six months after the audit announcements were mailed. Coefficients and standard errors are multiplied by 100 to express effects in percent. Robust standard errors in parentheses, clustered at the level of the audited firm.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

Table 7: Spillover Effects on Trading Partners' VAT Payments

8 Short summary

- The paper studies whether the VAT's paper trail helps tax enforcement.
- Chile's VAT creates records because firms want receipts for input deductions.
- The paper uses two randomized field experiments with administrative VAT data.
- The deterrence letter increases VAT payments, proving that the tax authority can shift perceived monitoring risk.
- The deterrence response is much stronger for final sales than for inter-firm transactions.
- This implies that the VAT paper trail had already reduced evasion on transactions between firms.
- The audit preannouncement creates spillovers to suppliers, but not clients.
- This supplier-side spillover shows the self-enforcing mechanism in action.
- The VAT paper trail is not self-enforcing by itself; it works through the interaction between information and credible deterrence.
- The results imply that audit policy should account for network multiplier effects.

9 Conclusion

9.1 Core conclusion

The paper provides direct micro-empirical evidence that the VAT facilitates tax enforcement through third-party information. Transactions covered by a VAT paper trail respond less to an increase in

perceived audit probability, consistent with lower ex ante evasion on those transactions. Enforcement also spills over to suppliers, showing that the paper trail propagates deterrence up the VAT chain.

9.2 Economic intuition

The VAT turns firms into information providers about one another. A buyer wants a receipt because the receipt lowers its own tax bill, but the same receipt reveals the seller's sales. This creates a paper trail that can discipline sellers. At the final consumer stage, this incentive disappears, so evasion is easier and deterrence has a larger effect.

9.3 Incremental contribution revisited

The paper's main incremental contribution is to move the VAT self-enforcement claim from theory and cross-country correlations to micro-randomized evidence. The Letter Message Experiment tests whether the paper trail already deters evasion in the full economy. The Spillover Experiment shows the mechanism by which monitoring one firm changes the behavior of its trading partners.

9.4 Implications for tax policy

- **Audit targeting.** Tax authorities should consider not only the direct yield from auditing a firm, but also the spillover effects through its trading network.
- **Final-stage enforcement.** Retail and final-consumer transactions are a weak point in the VAT because consumers lack deduction incentives.
- **Information systems.** Electronic invoicing, online billing, and digital receipts may increase the value of VAT-like systems by making cross-checks cheaper and more automatic.
- **Formalization.** If formal firms prefer trading with other formal firms because they can provide receipts, enforcing formalization at downstream stages may help formalize upstream chains.
- **Tax instrument choice.** VAT and retail sales taxes may be equivalent in simple models, but not when enforcement and information generation matter.

9.5 Limitations

- The Chilean Tax Authority is relatively capable, so effects may differ in countries where enforcement is weaker or corruption is higher.
- The Letter Message Experiment identifies responses to a perceived audit-risk change, not the full long-run effect of a permanent audit-policy change.
- The Spillover Experiment has attrition because trading partner information is collected only for audited firms.
- The paper studies formal-sector firms; the VAT may also affect incentives to remain informal.
- The experiments reveal compliance responses, but they do not fully quantify all real-output responses, long-run equilibrium effects, or welfare consequences.

9.6 Takeaway

The paper's central message is: **tax enforcement depends on information.** The VAT works better than otherwise equivalent taxes because it creates verifiable paper trails between firms. But information alone is not enough; the paper trail becomes powerful when firms believe that the tax authority can and will use it.